

Survey on Engineering Practices for LLM-Based Systems

Participant Information Sheet

The Karlsruhe Institute of Technology (KIT) and the University of Duisburg-Essen are conducting a research study on how software engineers design, implement, and maintain systems that integrate Large Language Models (LLMs), such as OpenAI GPT, Google Gemini, or Anthropic Claude.

The goal of this study is to better understand current practices, challenges, and needs in developing LLM-based software systems. In particular, we aim to investigate issues related to prompt design, interaction logic, reliability, and maintainability of such systems.

Your participation is voluntary. Before deciding whether to take part, please read this information carefully and take as much time as you need.

What does participation involve?

If you choose to participate, you will be asked to complete an online questionnaire. The survey includes questions about your experience with LLM-based systems, the challenges you encounter, and your perspective on possible improvements.

The questionnaire takes approximately **5–10 minutes** to complete.

Data collection and privacy

The survey is **anonymous**. No personally identifying information will be collected unless explicitly stated. All data will be processed in accordance with the **General Data Protection Regulation (EU) 2016/679 (GDPR)** and used exclusively for academic research purposes.

Risks and benefits

Participation in this study involves **no known risks**, as it consists solely of answering survey questions. While there are no direct benefits to individual participants, your responses will contribute to research aimed at improving tools, methods, and practices for engineering reliable and trustworthy AI-based software systems.

Voluntary participation

Your participation is entirely voluntary. You may choose not to participate or stop the survey at any time without any consequences.

Contact

If you have any questions regarding this study, please contact:

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Data Protection Notice

Read the informative sheet according to Art. 13 of the General Data Protection Regulation (EU) 2016/679 ("Regulation" or "GDPR")

<https://www.comp.kit.edu/english/datenschutz.php>

* Obligatoria

Screening & Background

1. Have you worked with LLMs in software systems? *

- ☐ Yes, in production systems
- ☐ Yes, prototypes/research
- ☐ No

2. What is your primary role? *

- ☐ Software engineer
- ☐ ML/AI engineer
- ☐ DevOps/MLOps
- ☐ QA/testing
- ☐ Researcher
- ☐ Altro

3. How many years of experience with LLM-based systems do you have? *

- ☐ < 1 year
- ☐ 1-2
- ☐ 3-5
- ☐ >5

4. Consider the development of LLM/AI-based software (GenAI-ware) *

	Highly unaware	Unaware	Knowledgeable	Highly knowledgeable
How knowledgeable are you about the topic?	☐	☐	☐	☐

Current practice

5. How do you implement LLM interactions today? (multiple choice) *

- ☐ Hardcoded prompts in code
- ☐ Prompt templates
- ☐ Frameworks (LangChain, Semantic Kernel, etc.)
- ☐ DSL-like tools (Guidance, Guardrails, etc.)
- ☐ Low-code platforms (n8n, etc.)
- ☐ Altro

6. Where are prompts/interaction logic handled? *

- ☐ Inline in application code
- ☐ Separate files/templates
- ☐ Mixed
- ☐ Altro

Pain Points

7. Do you agree with the following statements? *

	Strongly agree	Agree	Neither agree nor disagree	Disagree	Strongly disagree
"Prompt logic in my systems is difficult to understand and reason about."	☐	☐	☐	☐	☐
"Debugging LLM-based pipelines is difficult and time-consuming."	☐	☐	☐	☐	☐
"Invalid LLM outputs often propagate and break downstream components."	☐	☐	☐	☐	☐
"It is hard to enforce structure or constraints on LLM outputs."	☐	☐	☐	☐	☐
"Maintaining and evolving LLM interaction logic is challenging."	☐	☐	☐	☐	☐
"Separating deterministic logic from LLM behavior would help improve system quality."	☐	☐	☐	☐	☐

8. To demonstrate that you are reading carefully, please select 'Disagree' for this statement.

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
Do you agree?	☐	☐	☐	☐	☐

Desired Capabilities

9. How useful would the following features be in your work? (Higher values indicate higher usefulness) *

	1	2	3	4	5
Explicit representation of LLM interaction workflows	☐	☐	☐	☐	☐
Static analysis of interaction logic	☐	☐	☐	☐	☐
Ability to constrain outputs via schemas/grammars	☐	☐	☐	☐	☐
Built-in validation of outputs	☐	☐	☐	☐	☐
Clear separation between LLM calls and control logic	☐	☐	☐	☐	☐

Reaction to LLM Scripting Language Concept

10. Consider a DSL for defining LLM interactions as structured programs (with explicit context, constraints, and control flow), separate from general-purpose code (Higher values indicate higher usefulness). *

	1	2	3	4	5
How useful would such an approach be?	☐	☐	☐	☐	☐

11. Do you agree with following benefits of a DSL for defining LLM interactions as structured programs? *

	Strongly agree	Agree	Neutral	Disagree	Strongly disagree
Debugging will improve	☐	☐	☐	☐	☐
Reliability will improve	☐	☐	☐	☐	☐
Maintenance will be easier	☐	☐	☐	☐	☐
Testing and verification will improve	☐	☐	☐	☐	☐
No benefit is expected	☐	☐	☐	☐	☐
Finding bugs will be harder	☐	☐	☐	☐	☐

12. How concerned are you about the following aspects of a DSL for defining LLM interactions as structured programs? *

	Very unconcerned	Unconcerned	Indifferent	Concerned	Very concerned
Learning curve	☐	☐	☐	☐	☐
Integration with existing tools	☐	☐	☐	☐	☐
Performance overhead	☐	☐	☐	☐	☐
Limited flexibility	☐	☐	☐	☐	☐
Not needed	☐	☐	☐	☐	☐

Optional questions

13. Describe the most challenging issue you faced with LLM-based systems.

14. Would you consider adopting a DSL for defining LLM interactions as structured programs? Why / why not?

15. If you have other general comments and opinions on the topic, use them to share with us.

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